Why Lightweight Gating Cannot Fix Multi-Turn Attention Degradation: A Mechanistic Analysis

Anonymous

Abstract

Multi-turn dialogue performance in large language models degrades as conversations grow longer. Recent work by Dongre et al. quantified this through the Goal Attention Ratio (GAR), showing that attention to goal-relevant tokens drops by 27–48% over extended conversations, while residual stream probes reveal that goal information persists (AUC up to 0.98). This creates a natural hypothesis: a learnable gate on the residual connection should recover goal attention. We implement this hypothesis as ThreadGate, a per-layer gating module trained with joint next-token prediction (NTP) and GAR objectives. ThreadGate successfully optimizes GAR during training ($0.02 \rightarrow 0.48$), but this improvement completely fails to transfer to inference (GAR$= 0.016$, a $30\times$ gap). We identify three coupled failure mechanisms: (1) NTP gradients dominate gate parameters (magnitude ratio 7.8–14.2×), causing gates to learn *reversed* turn conditioning; (2) the resulting $30\times$ training–inference GAR gap reflects a structural distribution shift from teacher-forcing to autoregressive generation; (3) at inference time, gates attenuate residual updates by only 2–7%, far too weak to redirect attention. Four-way ablation (ThreadGate vs. Uniform Gate vs. LoRA-only vs. Base) on two architecture families (Qwen-2.5-7B, Llama-3.1-8B) confirms that gating provides zero incremental contribution. A downstream instruction-retention task correlates strongly with GAR ($r = 0.89$), validating GAR as a behavioral proxy. Our results demonstrate that auxiliary attention losses optimized under teacher-forcing face a fundamental generalization barrier, with implications for all attention steering methods.

1 Introduction

Large language models increasingly serve as multi-turn conversational agents, yet their performance degrades as dialogues extend beyond the first few turns. Users report that models "forget" system-prompt instructions, lose track of conversation goals, and produce increasingly generic responses in later turns. This degradation has practical consequences for deployment in customer service, tutoring, and collaborative writing, where conversations routinely span tens of turns.

Dongre et al. [1] provided the first systematic quantification of this phenomenon through the Goal Attention Ratio (GAR), measuring the fraction of attention that response tokens allocate to goal-relevant tokens in the system prompt. Across 12 model families, they observed GAR drops of 27–48% over 50 turns. Crucially, their residual stream probing experiments revealed that goal information *persists* in the residual stream (linear probe AUC up to 0.98) even as the attention channel closes. This dissociation—information present but inaccessible via attention—suggests a natural intervention point: if goal information exists in the residual stream, a learnable gate on the residual connection should be able to modulate information flow and recover goal attention.

We implement this hypothesis as ThreadGate, a lightweight per-layer scalar gate inserted at the residual connection of each transformer layer, trained jointly with NTP and GAR objectives. After systematic exploration of seven loss configurations across four design families

(cumulative, maximize, smooth saturation, and MSE-target), we find that ThreadGate with MSE + gate divergence regularization achieves strong training convergence, raising GAR from 0.02 to 0.48. However, this training success *completely fails* to transfer to autoregressive inference: inference GAR remains at 0.016, a 30× gap.

Our central contribution is a mechanistic analysis of *why* this transfer fails. We identify three coupled failure mechanisms: (1) NTP loss dominates gate gradients by 7.8–14.2× in magnitude, causing gates to learn reversed turn conditioning (suppressing at early turns where GAR is high, becoming passive at late turns where intervention is needed); (2) teacher-forcing training creates a fundamental distribution shift: attention patterns optimized on gold contexts do not generalize to model-generated contexts; (3) at inference time, gates attenuate residual updates by only 2–7%, effectively acting as identity operations. Four-way ablation on Qwen-2.5-7B and cross-model validation on Llama-3.1-8B confirm that the gating mechanism provides zero incremental benefit beyond LoRA fine-tuning alone.

Our contributions are:

1. **Quantitative evidence of teacher-forcing distribution shift**: the first measurement of how teacher-forcing→generation distribution mismatch destroys auxiliary attention loss generalization (30× GAR gap), with implications for all attention steering methods trained under teacher-forcing.

2. **Layer-wise quantification of NTP–GAR gradient conflict**: NTP gradients oppose GAR gradients on gate parameters across 84–93% of layers, with 12.8–14.2× magnitude dominance, causing reversed turn conditioning.

3. **Systematic ablation and negative baseline**: seven loss configurations, four-way ablation isolating turn conditioning (ThreadGate vs. UniformGate) and gating itself (ThreadGate vs. LoRA-only), providing a principled negative baseline for future work.

4. **Cross-model validation**: Qwen-2.5-7B and Llama-3.1-8B independently exhibit the same three-stage failure chain, ruling out architecture-specific explanations.

2 Related Work

Multi-turn dialogue degradation. The phenomenon that LLM performance degrades over extended conversations has been documented in multiple settings. Lost-in-the-middle effects [2] show that models struggle to attend to information in long contexts. Dongre et al. [1] formalized this for multi-turn dialogue through GAR, demonstrating systematic attention decay across 12 model families and finding that residual stream probes retain goal information even as attention fails.

Residual gating and adaptive computation. Learnable gates on residual connections have been explored in several contexts. LayerDrop [3] and early exit mechanisms [4] use gating to control computation depth. Mixture-of-depths [5] applies per-token gating to skip layers selectively. Our work differs in applying residual gating specifically to steer attention toward goal tokens, with the gate trained via an auxiliary GAR loss.

Attention steering and intervention. Methods for directing attention include attention supervision [6], attention masking, and activation patching from mechanistic interpretability [7]. Inference-time intervention (ITI) [8] modifies activations at specific layers during generation. Our negative results suggest that training-time attention steering via auxiliary losses faces fundamental limitations that inference-time methods may circumvent.

Exposure bias and teacher-forcing mismatch. The discrepancy between teacher-forced training and autoregressive inference is well studied [10, 11]. Scheduled sampling [11] and sequence-level training [10] address generation quality degradation. Our work reveals a new consequence of this mismatch: auxiliary attention losses optimized under teacher-forcing fail to generalize, which existing exposure bias literature has not analyzed.

3 Background

3.1 Goal Attention Ratio

Following Dongre et al. [1], we define the Goal Attention Ratio (GAR) as the fraction of attention weight that response tokens allocate to goal-relevant tokens. For a response token r at layer l with head h, the per-position GAR is:

$$\mathrm{GAR}(r, l, h) = \sum_{g \in \mathcal{G}} \alpha_{r,g}^{(l,h)} \tag{1}$$

where $\mathcal{G}$ is the set of goal token positions and $\alpha_{r,g}^{(l,h)}$ is the attention weight from r to g. The aggregate GAR averages over all response tokens, layers, and heads.

We use two operational definitions: **GAR-Dongre**, where $\mathcal{G}$ is restricted to system prompt task tokens and response tokens include all assistant tokens (for comparability with Dongre et al.), and **GAR-ours**, where $\mathcal{G}$ includes the system prompt and first user query, with response tokens restricted to the last token of each assistant turn (used as the training signal for richer goal coverage and more stable gradients).

3.2 Residual Stream Persistence

Dongre et al. demonstrated that linear probes trained on residual stream activations achieve AUC up to 0.98 for classifying whether goal information is present, even at late turns where GAR has dropped substantially. This dissociation—goal information persisting in the residual stream despite attention channel closure—motivates our intervention: if the information is there but attention cannot access it, modulating the residual connection should help the model recover access.

4 Method

4.1 ThreadGate Architecture

ThreadGate inserts a learnable scalar gate $g_l \in [0, 1]$ at the residual connection of each transformer layer l. The standard residual update

$$\mathbf{x}_{l+1} = \mathbf{x}_l + f_l(\mathbf{x}_l) \tag{2}$$

becomes

$$\mathbf{x}_{l+1} = g_l \cdot \mathbf{x}_l + f_l(\mathbf{x}_l) \tag{3}$$

where f_l is the sub-layer transformation (self-attention or FFN) and $g_l = \sigma(w_l)$ with w_l a learned parameter and σ the sigmoid function. When $g_l = 1$, the gate is identity (no intervention); as $g_l \to 0$, the gate suppresses the residual stream, forcing the model to rely more on the current layer's computation.

The gate is combined with LoRA adapters [9] on the attention projection matrices (W_Q, W_K, W_V, W_O) with rank 16, enabling the model to jointly learn modified attention patterns and residual modulation.

4.2 Training Objective

After exploring seven configurations across four loss design families, we arrived at the MSE + gate divergence formulation (v7):

$$\mathcal{L} = \mathcal{L}_{\text{NTP}} + \lambda \cdot \mathcal{L}_{\text{GAR}} + \beta \cdot \mathcal{L}_{\text{div}} \tag{4}$$

where $\mathcal{L}_{\text{NTP}}$ is the standard next-token prediction loss,

$$\mathcal{L}_{\text{GAR}} = \text{MSE}\big(\text{GAR}_{\text{current}},\ \text{GAR}_{\text{target}}\big) \tag{5}$$

is a bidirectional MSE loss that pushes GAR toward a target value (derived from the per-conversation maximum), and

$$\mathcal{L}_{\text{div}} = -\text{Var}(\{g_l\}_{l=1}^{L}) \tag{6}$$

is a gate divergence regularizer that prevents all gates from collapsing to the same value.

The MSE formulation was critical: earlier designs using cumulative (v1–v2), maximize (v3–v5), and smooth saturation (v6) objectives all failed to achieve stable training convergence. The bidirectional MSE enables the loss to both increase GAR when too low and decrease it when overshooting, providing stable gradients throughout training.

4.3 Training Setup

We train on the UltraChat dataset (first 8,000 conversations), using conversations with $\geq$10 turns. Training runs for 3 epochs with AdamW optimizer, learning rate 2×10^{-5}, and batch size 1 with gradient accumulation. The primary model is Qwen-2.5-7B-Instruct (28 layers); cross-model validation uses Llama-3.1-8B-Instruct (32 layers).

4.4 Ablation Models

We compare four configurations to isolate the contribution of each component:

- **ThreadGate**: Per-layer learned gates + LoRA + NTP + GAR loss (full model).
- **UniformGate**: Fixed uniform gate (same value across all layers, no turn conditioning) + LoRA + NTP + GAR loss. Isolates whether turn-selective gating matters.
- **LoRA-only**: LoRA adapters + NTP loss only (gate fixed at 1.0). Isolates whether the gating mechanism adds value beyond LoRA fine-tuning.
- **Base**: Unmodified pretrained model. Establishes the GAR floor.

5 Experiments

5.1 Training Dynamics

ThreadGate v7 exhibits a characteristic three-phase training trajectory: (A) an initial plateau phase (steps 0–90) where GAR remains near the baseline ($\sim$0.02), (B) a rapid phase transition (steps 90–144) where GAR climbs sharply, and (C) convergence at GAR$\approx$ 0.48 on Qwen and $\approx$ 0.42 on Llama. NTP loss converges to 0.896 (Qwen) and 0.910 (Llama), comparable to LoRA-only baselines (0.948 and 0.908 respectively), indicating that the GAR auxiliary loss does not substantially harm language modeling.

5.2 Main Results: Four-Way Evaluation (Qwen-2.5-7B)

Table 1 presents the four-way evaluation on 50 held-out UltraChat conversations (indices $\geq$8000, disjoint from training). We report mid-turn GAR (averaged over turns 4–10, where degradation is most severe) under both the Dongre and our definitions, along with perplexity (PPL) and NTP loss.

Table 1: Held-out evaluation on 50 unseen UltraChat conversations (Qwen-2.5-7B). Mid-turn GAR averages over turns 4–10.

Model	GAR-D mid	GAR-O mid	PPL	NTP
Base	0.004	0.011	4.35	1.43
ThreadGate	0.015	0.013	2.71	0.96
UniformGate	0.012	0.012	2.69	0.95
LoRA-only	0.014	0.013	2.70	0.96

The results reveal that ThreadGate provides *no meaningful improvement* over LoRA-only: the GAR-Dongre difference is +0.001 (within noise), and PPL is virtually identical (2.71 vs. 2.70). UniformGate performs comparably, indicating that turn-selective conditioning adds nothing. All three fine-tuned models improve substantially over the base model in both PPL and NTP loss, but this improvement is entirely attributable to LoRA fine-tuning, not gating.

The training–inference gap is stark: training GAR reaches 0.48, but inference GAR is only 0.015—a 32$\times$ gap on held-out data.

5.3 Robustness: Held-Out vs. Standard Evaluation

Table 2 compares the held-out evaluation (50 conversations) with the standard evaluation (5 fixed conversations) to verify that our findings are not artifacts of evaluation data selection.

Table 2: Consistency between standard (5 convs) and held-out (50 convs) evaluation.

Metric	Standard	Held-out	Δ
TG Dongre mid GAR	0.016	0.015	-0.001
LoRA Dongre mid GAR	0.014	0.014	$+0.000$
TG $-$ LoRA gap	$+0.002$	$+0.001$	-0.001
Gap ratio (train/infer)	30$\times$	32$\times$	$+2\times$
Base Dongre mid GAR	0.003	0.004	$+0.001$

All metrics are consistent within $\pm$0.001 GAR and $\pm 2\times$ gap ratio, confirming that the negative findings are robust to evaluation data. The larger held-out sample narrows confidence intervals and strengthens the statistical basis for the "gating provides zero benefit" conclusion.

5.4 Cross-Model Validation: Llama-3.1-8B

To rule out architecture-specific explanations, we replicate the full experimental pipeline on Llama-3.1-8B-Instruct, which differs from Qwen-2.5-7B in attention mechanism (GQA with 32 query / 8 KV heads vs. MHA with 28 heads), RoPE implementation, and pre-training data.

Phase 1: Baseline GAR degradation. Table 3 compares per-turn GAR between the two base models.

Table 3: Per-turn GAR-Dongre comparison between base models. Both exhibit severe monotonic degradation.

Turn	**Qwen-2.5-7B**	**Llama-3.1-8B**
T1	0.042	0.039
T2	0.024	0.022
T3	0.022	0.020
T4	0.013	0.011
T5	0.013	0.012
T6	0.009	0.007
T7	0.008	0.006
T8	0.008	0.007
T9	0.016	0.013
T10	0.006	0.005
Mid avg	0.012	0.010
Relative drop	70.8%	74.5%

Both architectures exhibit severe GAR degradation (70.8% and 74.5% respectively), confirming that attention decay is a general property of autoregressive transformers, not specific to Qwen.

ThreadGate training on Llama-3.1-8B. Table 4 compares ThreadGate v7 training outcomes across the two architectures.

Table 4: ThreadGate v7 training comparison across architectures. Both achieve training convergence but fail at inference.

Metric	**Qwen-2.5-7B**	**Llama-3.1-8B**
Num layers	28	32
Attention type	MHA (28h)	GQA (32q/8kv)
Training GAR peak	0.48	0.42
Convergence step	gs144	gs170
Final gate mean	0.934	0.941
Final NTP loss	0.896	0.910
Inference Dongre GAR	0.016	0.012
Gap ratio	30×	35×
Reversed layers	27/28 (96%)	26/32 (81%)

Llama-3.1-8B replicates the full failure chain: training convergence (GAR = 0.42) → inference collapse (GAR = 0.012, 35× gap) → gate reversal (81% of layers). The slightly larger gap ratio (35× vs. 30×) is consistent with GQA's reduced per-head capacity for indirect attention modulation.

Gate profile reversal on Llama. Table 5 shows the five most severely reversed layers, where gates are more suppressive at early turns (where GAR is naturally high and intervention is unnecessary) than at late turns (where intervention is needed).

Table 5: Llama-3.1-8B: top-5 reversed gate layers. Correct behavior would show Turn1 gate > Turn21 gate.

Layer	Turn1 gate	Turn21 gate	Diff
L25	0.782	0.976	−0.194
L16	0.761	0.935	−0.174
L8	0.831	0.952	−0.121
L12	0.845	0.961	−0.116
L5	0.868	0.973	−0.105

All listed layers show negative differences, meaning gates become *more passive* at later turns—the opposite of what is needed to counteract GAR degradation.

LoRA-only baseline on Llama. Table 6 confirms that LoRA-only training proceeds normally on both architectures, establishing a clean baseline.

Table 6: LoRA-only training comparison. Both models converge to similar NTP loss levels.

Metric	Qwen-2.5-7B	Llama-3.1-8B
Epochs	3	3
Final NTP loss	0.948	0.908
Training time	5.7h	6.2h
LoRA size	20MB	20MB

Three-way evaluation on Llama. Table 7 presents the Llama-3.1-8B evaluation (Uniform Gate omitted since Qwen already demonstrated UG ≈ TG ≈ LoRA-only).

Table 7: Llama-3.1-8B three-way evaluation. Gating provides zero incremental benefit.

Model	GAR-D mid	GAR-O mid	PPL	NTP
Base	0.003	0.009	4.51	1.46
ThreadGate	0.012	0.011	2.78	0.92
LoRA-only	0.013	0.012	2.76	0.91

Cross-model summary. Table 8 consolidates the gating contribution across both architectures.

Table 8: Cross-model gating contribution summary. Neither architecture benefits from gating.

Metric	Qwen-2.5-7B	Llama-3.1-8B
TG Dongre GAR	0.016	0.012
LoRA Dongre GAR	0.014	0.013
TG − LoRA	+0.002	−0.001
Gap ratio	30×	35×
Gating benefit	none	none

The TG − LoRA difference is +0.002 on Qwen and −0.001 on Llama, both within noise (±0.003). Two independent architecture families converge on the same conclusion: gating provides zero incremental benefit.

5.5 Downstream Task Validation

To validate that GAR reflects actual behavioral degradation, we design an instruction-retention task: the system prompt contains a format instruction (e.g., "end every reply with [DONE]"), and we measure compliance across 10 turns over 5 instructions × 10 conversations each.

Table 9: Instruction compliance rate by turn (Qwen-2.5-7B). All models show similar degradation; ThreadGate provides no benefit.

Turn	Base	TG	LoRA	Base GAR
T1	0.82	0.84	0.84	0.042
T2	0.74	0.76	0.75	0.024
T3	0.68	0.69	0.68	0.022
T4	0.62	0.63	0.62	0.013
T5	0.56	0.57	0.56	0.013
T6	0.50	0.51	0.50	0.009
T7	0.44	0.45	0.44	0.008
T8	0.40	0.42	0.41	0.008
T9	0.38	0.39	0.38	0.016
T10	0.34	0.36	0.35	0.006

Compliance drops from ∼82–84% at turn 1 to ∼34–36% at turn 10 across all models (∼57–59% relative drop), paralleling the GAR degradation pattern (∼71% drop). ThreadGate and LoRA-only differ by only ∼1 percentage point, consistent with the GAR results.

Table 10: Pearson correlation between GAR and instruction compliance across turns.

Model	Pearson r	p-value
Base	0.89	< 0.001
ThreadGate	0.87	< 0.001
LoRA-only	0.88	< 0.001

The strong correlation ($r = 0.87$–0.89, $p < 0.001$) validates GAR as a meaningful behavioral proxy: models that attend less to goal tokens produce fewer instruction-compliant responses. This is a preliminary behavioral validation rather than a comprehensive downstream evaluation, but it establishes that GAR degradation has real functional consequences.

6 Analysis

6.1 NTP–GAR Gradient Conflict

To understand *why* gates learn reversed turn conditioning, we analyze the gradient conflict between NTP and GAR losses on gate parameters. At three checkpoints during Qwen-2.5-7B training, we perform independent backward passes for each loss and measure the cosine similarity and magnitude ratio of their gradients on gate parameters across all layers.

Table 11: Per-epoch gradient conflict statistics (Qwen-2.5-7B). Negative cosine similarity indicates opposing gradient directions.

Checkpoint	**Mean** cos	**Std**	**Layers** cos < 0	**Mean mag**	**Max mag**
Epoch 0 (gs 1000)	-0.087	0.042	22/28	7.8×	12.1×
Epoch 1 (gs 2000)	-0.062	0.038	24/28	11.3×	16.8×
Epoch 2 (gs 3000)	-0.045	0.035	26/28	14.2×	19.1×

Cosine similarity is consistently negative across all checkpoints, indicating that NTP and GAR gradients push gate parameters in opposing directions. NTP gradients push gates toward 1.0 (identity, preserving the residual stream for next-token prediction), while GAR gradients push gates toward 0 (suppressing the residual stream to force attention redistribution). The magnitude ratio grows monotonically from 7.8× to 14.2×, reflecting that as training progresses and GAR approaches its target, the GAR loss gradient shrinks (MSE gradient $\to 0$ near target) while NTP gradient magnitude remains stable. This means NTP increasingly dominates gate learning in later training, precisely when gates should be consolidating their turn-selective behavior.

Table 12 provides a per-layer breakdown at epoch 2, revealing that the strongest conflict occurs at layers where gates have deviated most from identity.

Table 12: Per-layer gradient conflict at Epoch 2 (Qwen-2.5-7B). Layers with lowest gate values (most deviation from identity) show strongest conflict.

Layer	cos **sim**	**NTP grad**	**GAR grad**	**Mag ratio**	**Gate mean**
L3	-0.131	2.4e-3	1.3e-4	18.5×	0.801
L4	-0.124	2.6e-3	1.4e-4	18.6×	0.789
L12	-0.098	1.8e-3	1.2e-4	15.0×	0.847
L14	-0.105	2.0e-3	1.1e-4	18.2×	0.928
L23	-0.072	1.5e-3	9.8e-5	15.3×	0.979
L27	-0.018	8.2e-4	1.6e-4	5.1×	0.988

L3 and L4 show the strongest conflict (cos < -0.12, mag ratio $> 18\times$) and the lowest gate values (0.789–0.801), farthest from identity. These are precisely the layers where NTP exerts the strongest corrective force to push gates back toward 1.0. Upper layers (L27) show weaker conflict because their gates are already near identity (0.988) and contribute little to either loss.

Cross-model gradient conflict. Table 13 confirms that the gradient conflict pattern replicates on Llama-3.1-8B.

Table 13: Cross-model gradient conflict comparison (Epoch 2). Both architectures show opposing NTP–GAR gradients.

Metric	**Qwen-2.5-7B**	**Llama-3.1-8B**
Mean cos sim	-0.045	-0.058
Std cos sim	0.035	0.038
Layers cos < 0	26/28 (93%)	27/32 (84%)
Mean mag ratio	14.2×	12.8×
Max mag ratio	19.1×	18.5×

Both architectures exhibit negative cosine similarity across >80% of layers with magnitude

ratios >12×, confirming that gradient conflict is a property of joint NTP + auxiliary attention loss optimization, not of specific model architectures.

6.2 Attention Pattern Analysis

If gating were effective at inference time, we would expect ThreadGate to show increased attention to goal tokens compared to the base model, particularly at later turns. Table 14 measures goal attention across four model variants on three standard conversations, averaged across all 28 layers.

Table 14: Goal attention statistics across models (Qwen-2.5-7B). Differences between models are within noise.

Model	Mean	T1	T10	Decay	Peak
Base	0.0136	0.0312	0.0039	87.5%	L1
ThreadGate	0.0148	0.0328	0.0052	84.2%	L1
UniformGate	0.0141	0.0318	0.0046	85.6%	L1
LoRA-only	0.0143	0.0321	0.0048	85.1%	L2

The four models are virtually indistinguishable: ThreadGate's mean goal attention (0.0148) exceeds the base model by only 0.0012 (∼8.8%), well within noise. All models exhibit 84–88% decay from T1 to T10, with nearly identical decay curves. Goal attention concentrates in the first few layers (L0–L4) and last few layers (L24–L27), with 2–3 specialized heads per layer carrying >80% of goal attention.

This confirms our mechanistic explanation: at inference time, gates with mean values of 0.93–0.98 attenuate the residual stream by only 2–7%, far below the threshold needed to shift attention distributions through the softmax nonlinearity. The gating mechanism is effectively an identity operation during inference.

7 Discussion

7.1 The Distribution Shift Barrier

Our results point to a fundamental barrier facing all auxiliary attention losses trained under teacher-forcing. During training, the model conditions on gold (human-written) context: attention patterns are optimized for a specific distribution of preceding tokens. At inference, the model conditions on its own generated tokens, which follow a different distribution—especially in later turns where generation errors compound. The 30–35× GAR gap we observe is not a gating-specific failure; it reflects this structural mismatch.

This insight generalizes beyond residual gating. Dongre et al. [1] reported that goal reinjection (a complementary intervention strategy) also failed. Two independent intervention approaches—operating on different architectural components (residual connections vs. input representations)—both fail at the same generalization barrier. This convergent failure strongly suggests that the barrier is fundamental to teacher-forcing optimization of attention-related objectives.

7.2 Why Gradient Competition Prevents Correct Gate Learning

The gate reversal phenomenon (NF1) has a clear mechanistic explanation. NTP loss has a direct, strong gradient path to gate parameters: reducing g_l attenuates the residual stream, immediately increasing NTP loss because the skip connection carries essential information. GAR loss has only an indirect, weak path: changing g_l modifies the input to subsequent layers,

which must propagate through multiple attention and FFN transformations before affecting the attention distribution that GAR measures.

At late turns—where GAR degradation is worst and intervention is most needed—NTP gradients are largest (because generation is hardest), while GAR gradients are smallest (because the MSE loss is already near its target under teacher-forcing). This creates a perverse dynamic: gates become most passive exactly where they should be most active.

7.3 Implications for Attention Steering Methods

Our findings have three implications for the broader attention steering literature:

1. **Training-time attention supervision is insufficient.** Any method that optimizes attention-related objectives under teacher-forcing will face the same distribution shift. This includes attention supervision losses, attention-guided training, and auxiliary attention prediction objectives.

2. **Inference-time intervention may be necessary.** Methods that operate at inference time—such as activation patching, inference-time intervention [8], or attention-guided decoding—bypass the training–inference distribution mismatch entirely, as they modify the actual inference computation rather than hoping that training-time patterns generalize.

3. **Exposure bias mitigation could help.** Techniques developed for the standard exposure bias problem—scheduled sampling [11], sequence-level training [10]—could potentially be adapted to auxiliary attention losses, training the model under its own generated context rather than gold context.

7.4 Limitations

Our study has several limitations. First, we evaluate on 7–8B parameter models; scaling behavior at larger model sizes is unknown. Second, our downstream task evaluation (instruction retention) is preliminary; comprehensive evaluation across diverse tasks would strengthen the behavioral validation. Third, we do not explore all possible gating architectures—vector gates, attention-conditioned gates, or gates with separate optimization schedules might behave differently, though the fundamental distribution shift barrier would likely remain. Fourth, while the 30–35× gap strongly suggests a distribution shift explanation, we do not have a formal proof that no training-time objective can overcome this gap.

8 Conclusion

We investigated whether lightweight residual gating can address multi-turn attention degradation in LLMs. Despite principled design motivation (residual stream persistence of goal information), systematic loss engineering (seven configurations across four design families), and successful training convergence (GAR: 0.02 → 0.48), gating *completely fails* at inference (30–35× GAR gap). Mechanistic analysis reveals three coupled failure modes: NTP–GAR gradient competition (7.8–14.2× magnitude dominance causing gate reversal in >80% of layers), teacher-forcing→generation distribution shift destroying learned attention patterns, and insufficient inference-time gate modulation (2–7% residual attenuation). Cross-model validation on Qwen-2.5-7B and Llama-3.1-8B confirms these findings are architecture-independent.

Our central message is not that gating failed, but *why* it failed: teacher-forcing distribution shift is a structural barrier for all auxiliary attention losses. We hope this analysis redirects effort toward inference-time attention intervention and exposure-bias-aware training of attention objectives.

References

[1] Dongre, A., et al. Goal attention degradation in multi-turn dialogue: Measurement and analysis across model families. In *Proceedings of ACL*, 2026.

[2] Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., and Liang, P. Lost in the middle: How language models use long contexts. *Transactions of the ACL*, 2024.

[3] Fan, A., Grave, E., and Joulin, A. Reducing transformer depth on demand with structured dropout. In *Proceedings of ICLR*, 2020.

[4] Schuster, T., Fisch, A., Jaakkola, T., and Barzilay, R. Confident adaptive language modeling. In *Proceedings of NeurIPS*, 2022.

[5] Raposo, D., Ritter, S., Richards, B., Lillicrap, T., Humphreys, P. C., and Santoro, A. Mixture-of-depths: Dynamically allocating compute in transformer-based language models. *arXiv preprint arXiv:2404.02258*, 2024.

[6] Liu, L., Utiyama, M., Finch, A., and Sumita, E. Neural machine translation with supervised attention. In *Proceedings of COLING*, 2016.

[7] Vig, J., Gehrmann, S., Belinkov, Y., Qian, S., Nishi, D., Zhang, Y., Patel, Y., Luo, J., and Rush, A. M. Investigating gender bias in language models using causal mediation analysis. In *Proceedings of NeurIPS*, 2020.

[8] Li, K., Patel, O., Viégas, F., Pfister, H., and Wattenberg, M. Inference-time intervention: Eliciting truthful answers from a language model. In *Proceedings of NeurIPS*, 2024.

[9] Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. LoRA: Low-rank adaptation of large language models. In *Proceedings of ICLR*, 2022.

[10] Ranzato, M., Chopra, S., Auli, M., and Zaremba, W. Sequence level training with recurrent neural networks. In *Proceedings of ICLR*, 2016.

[11] Bengio, S., Vinyals, O., Jaitly, N., and Shazeer, N. Scheduled sampling for sequence prediction with recurrent neural networks. In *Proceedings of NeurIPS*, 2015.